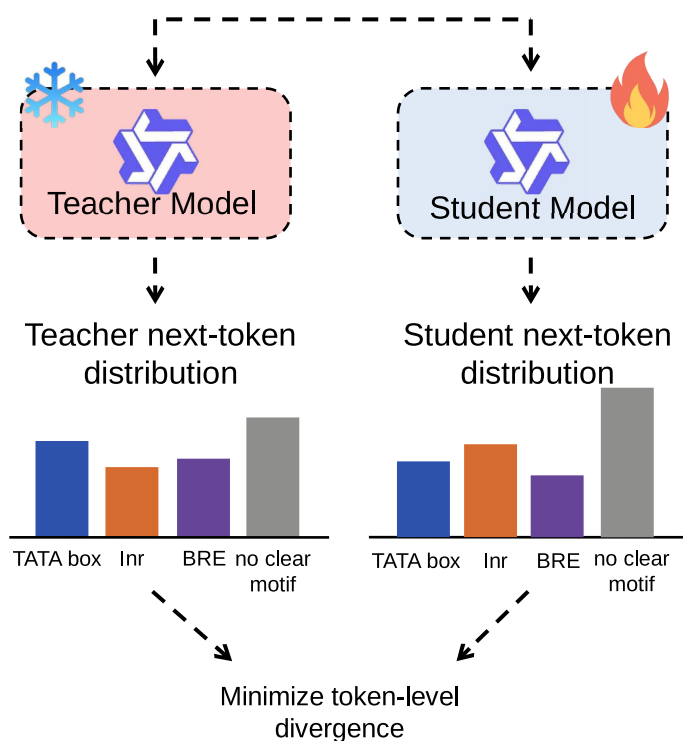


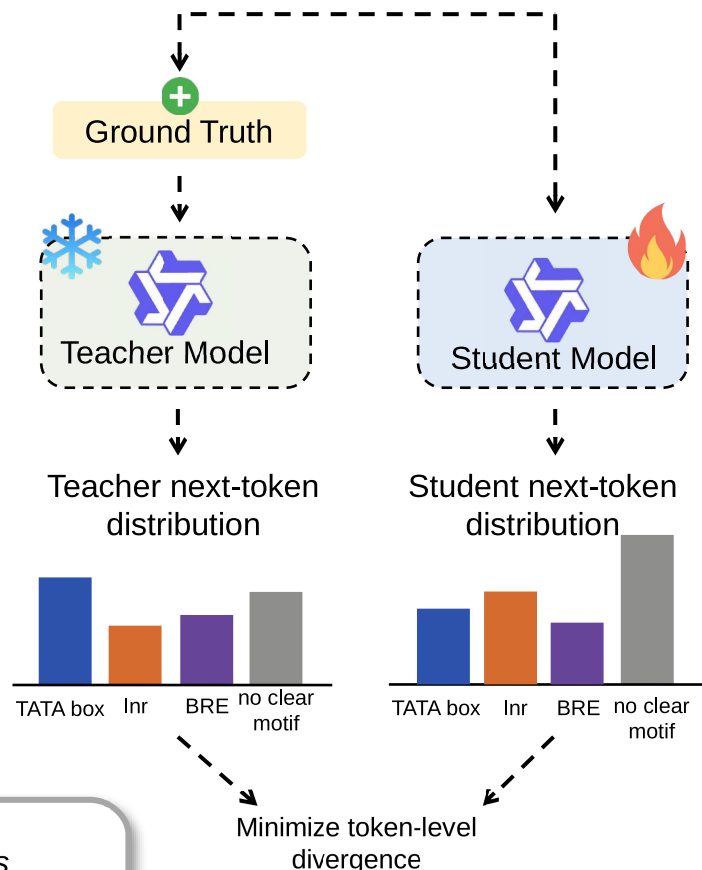
Problem: Does this DNA sequence contain a core promoter? 5'-TAGTTC[TTCC][TATATTG]TGAGGACTT...-3'

Student prefix: "The strongest promoter signal is ..."

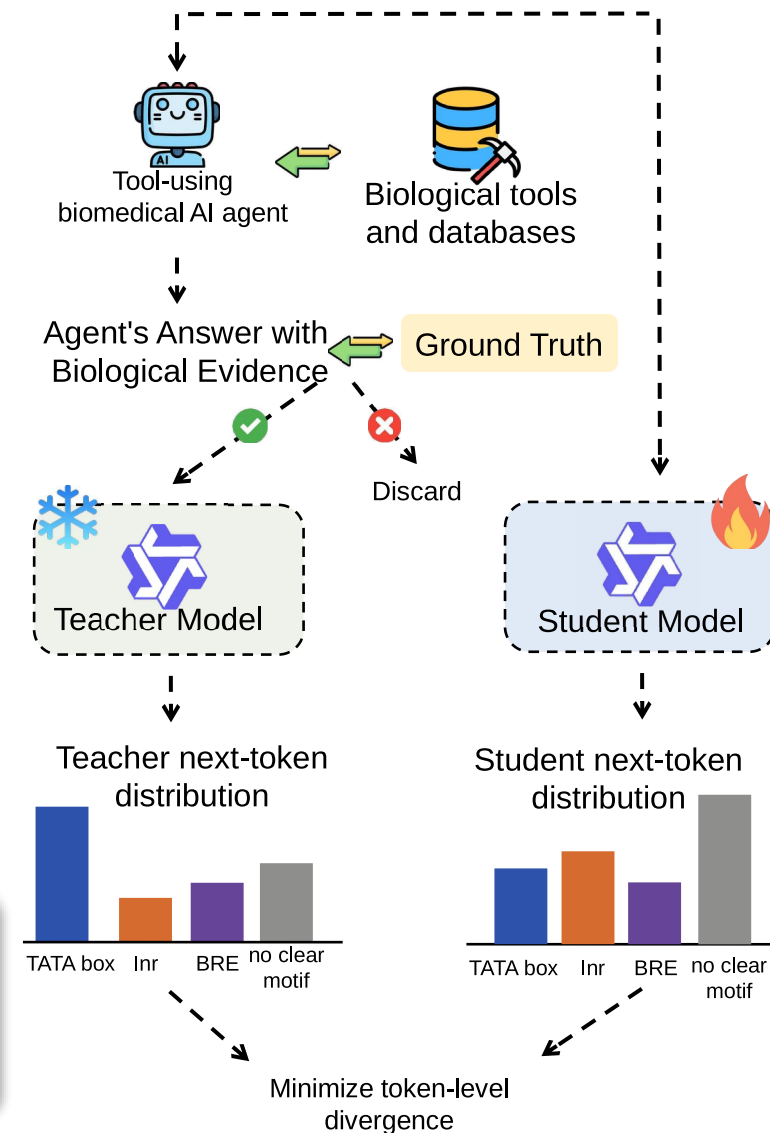
(A) OPD



(B) OPSD



(C) TA-OPD



Schematic motif-label probabilities

External teacher: frozen; typically larger model providing token-level supervision

Self-distillation teacher: frozen; same initialization, with privileged information

Student model: trainable; learns from the teacher's token distribution